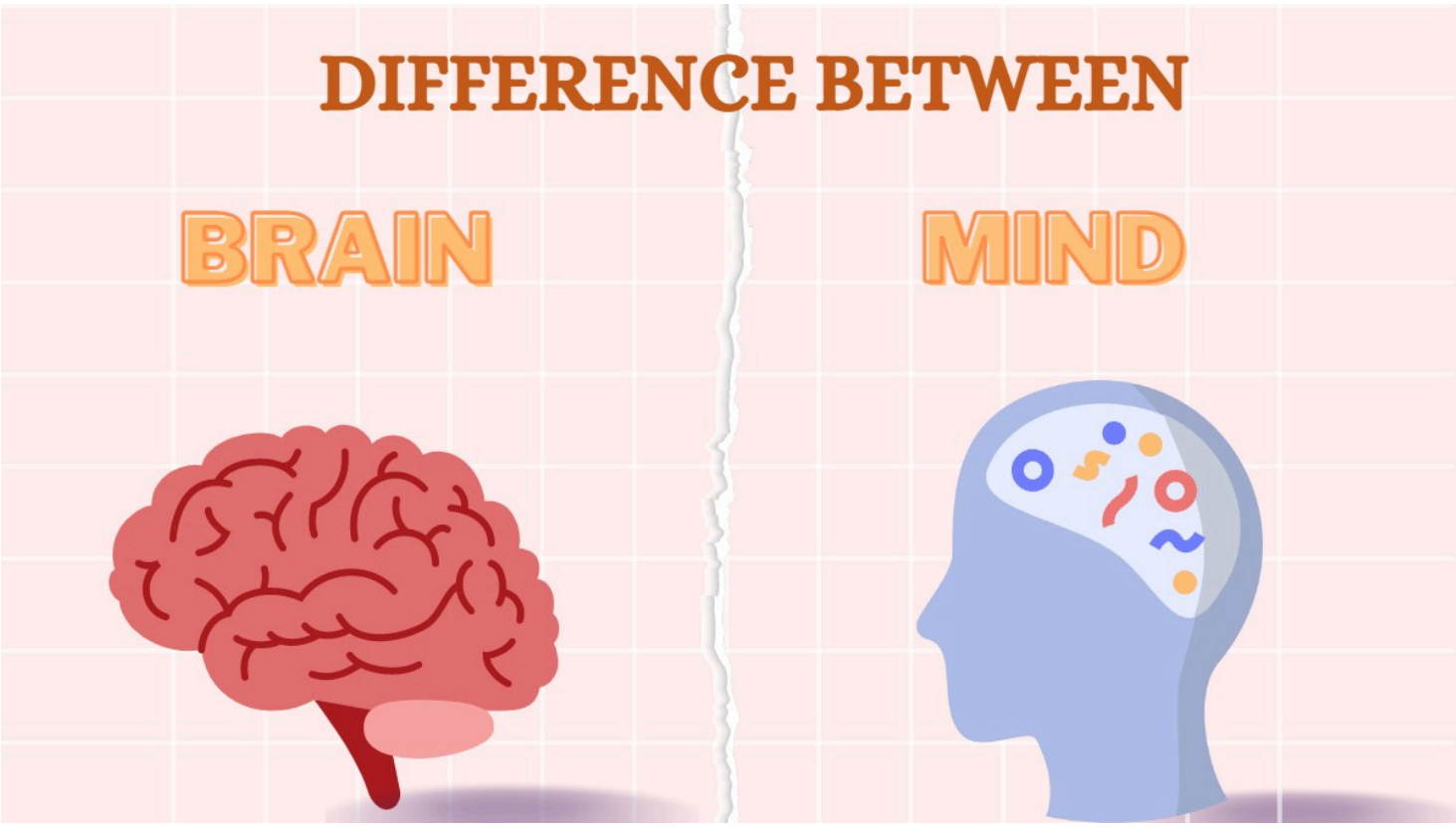
1. Brain vs Mind



Brain

- **Brain is a physical biological organ** inside the skull. It contains billions of neurons that communicate through **electrical and chemical** signals.
- It controls:
  - movement
  - emotions
  - thinking
  - memory
  - decision making.

Mind

- The **Mind** refers to the set of mental processes that arise from brain activity.
- These processes includes:
  - thinking
  - perception
  - imagination
  - reasoning
  - emotions
  - awareness

So, the **brain** is biological hardware, while the **mind** is the mental activity produced by that hardware.
Example: - When you solve a math problem:

- neurons in the brain fire
- the mind performs reasoning

Brain vs Mind (Comparison)

Brain	Mind
Physical organ	Mental processes
Made of neurons	Made of thoughts
Can be seen and studied biologically	Cannot be seen directly
Part of the body	Product of brain activity

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2. Consciousness

Definition

Consciousness is the **state of being aware of ourselves and our surroundings.**
It includes awareness of:

- thoughts
- feelings
- sensations

- environment
  Consciousness allows humans to:
- make decisions
- control actions
- understand experiences

Levels of Consciousness

Level	Meaning
Conscious	Fully aware and alert
Subconscious	Information stored but not actively aware
Unconscious	No awareness (sleep, coma)

Example: -
While driving:

- You **see traffic signals**
- You **hear sounds**
- You **make decisions**
  All of this happens through **conscious awareness**.

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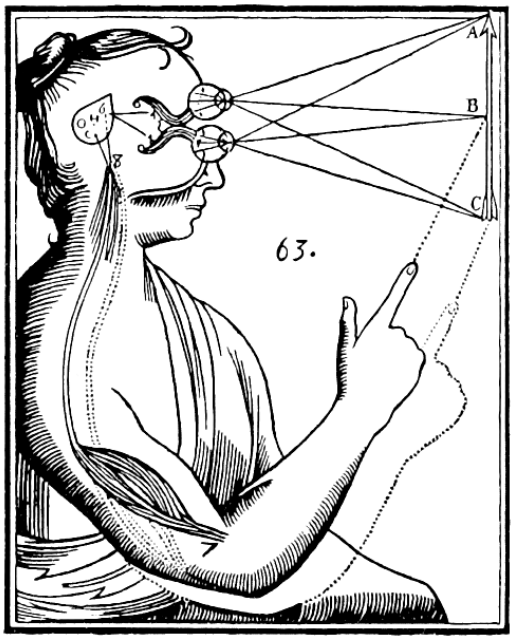
3. Problems Related to Mind (Mind–Body Problem)



The **Mind–Body Problem** is a **philosophical question**:

> How do **mental processes (mind)** interact with the **physical brain and body**?

In other words:

> How do thoughts, emotions and consciousness arise from brain activity?

This question has been debated for centuries in **philosophy, psychology, neuroscience, and cognitive computing**.
To explain this, philosophers gave different theories-

1. Dualism

Dualism says:

- Mind and body are two **separate entities**.
- Mind is **non-physical**.
- Brain and body are **physical**.
  Famous Philosopher: - **René Descartes (1596–1650)**
  Example:
- Your thoughts are different from your physical body.
- Thinking and physical movement are different substances.

2. Physicalism (Materialism)

Physicalism says:

- Mind and body are not two **separate entities**.
- Mind is a **product of brain activity**.
- Consciousness arises from neurons and brain processes.
- No separate 'Soul' substance required.
  Mental States = brain activity
  Example:
  When neurons fire in the brain → **thoughts and emotions occur**.

3. Functionalism

Functionalism says:

- Mental states are defined by **what they do, not by what they are made of**.
  This means:
- A mental process (mind) is identified by its function or role in a system, not by material.

Example:

Pain is defined not by brain chemistry but by its function:

- detecting damage
- causing reaction
- producing avoidance behavior

Functionalism suggests that any system performing the same function (process information) could have a mind, including computers.
This idea strongly influenced **Artificial Intelligence** and **Cognitive Computing**.
Example:

Conclusion

Theory	Main Idea
Dualism	Mind and body separate
Physicalism	Mind = brain activity
Functionalism	Mind = information processing function

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4. Relationship Between Consciousness and Cognitive Computing

Cognitive computing tries to **simulate some functions of consciousness**, but it **does not create real consciousness**.

Human Consciousness vs Cognitive Systems

Human Consciousness	Cognitive Computing
Real awareness	Simulated awareness
Has emotions	No real emotions
Self-aware	Limited self-monitoring
Experiences feelings	Processes data

Example:

When a **human doctor** diagnoses a disease:

- Understands symptoms
- Uses experience
- Feels empathy

When an **AI medical system** diagnoses:

- Analyzes medical data
- Uses machine learning models
- Suggests possible diseases

The system **acts intelligently**, but it is **not conscious like a human**.

Why Consciousness is Important in Cognitive Computing

Researchers study consciousness to build systems that can:

- understand human behavior
- interact naturally with humans
- improve decision making

- process complex information

Examples:

- Virtual assistants
- Chatbots
- Medical decision systems
- Self-driving cars

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5. Cognitive Workload?

Cognitive workload is the **amount of mental effort required to perform a task**.
Different tasks require different levels of mental processing

Levels of Cognitive Workload

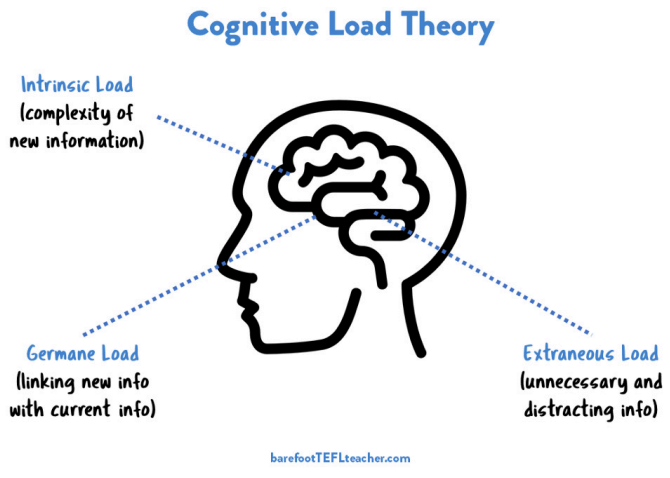
Situation	Level of Cognitive Workload
Watching TV	Low
Reading a book	Medium
Solving a math problem	High
Airplane pilot landing aircraft	Very High

- If Cognitive Workload becomes too high, performance decreases.
- The more demanding a task is, the higher workload required to perform it effectively.
- When task is complex, then more cognitive workload required.

Factor Influencing Cognitive Workload

- task complexity
- prior knowledge
- distractions

Types of Cognitive Workload



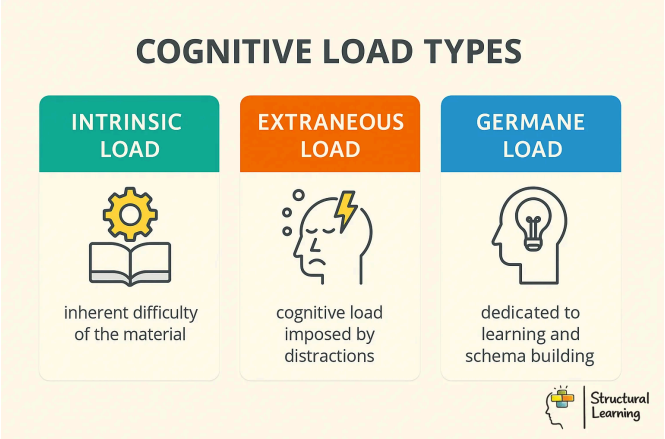
1. Intrinsic Cognitive Load

Intrinsic cognitive load is the **mental effort required because of the complexity of the task or information itself.**
intrinsic = things that cannot be regenerated/ changed easily

Example

Task	Intrinsic Load
2 + 2	Very low
Algebra equation	Medium
Quantum physics problem	Very high

2. Extraneous Cognitive Load



Extraneous cognitive load is the **extra mental effort caused by the way of information is presented (poor design or unnecessary information).**
extrinsic = things that can be regenerated/ modified

Example

Situation	Extraneous Load
Poorly designed website	High
Confusing lecture slides	High
Clear diagrams and explanation	Low

- If teacher poorly explain the complex topic, then the student do extra mental effort.

3. Germane Cognitive Load

Germane cognitive load is the **mental effort used to understand and learn new information and building knowledge structures (schemas).**

Example

Situation	Germane Load
Understanding concept	High
Creating mental connections	High
Memorizing patterns	Medium

When student:

- understand concept
- solve examples
- making connections between them
  then the Brain makes a knowledge structure, this is Germane load.

Comparison of Three Types

Type	Source	Can it be changed?	Example
Intrinsic	Task complexity	Difficult to change	Hard math problem
Extraneous	Presentation of information	can be reduced	Confusing slides
Germane	Learning process	can be reduced	Concept understanding

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6. Measuring Cognitive Workload

Measuring cognitive workload means **finding out how much mental effort a person is using during a task.**
Researchers measure it in different ways.

A. Physiological Methods (Body Signals)

Scientists observe body signals such as:

- **Heart rate**
- **Brain activity (EEG)**
- **Eye movement**
- **Pupil dilation**
  Example: - If a person is solving a hard problem, **brain activity increases**.

B. Performance-Based Methods

Measure how well a person performs a task.
Indicators include:

- task completion time
- number of mistakes

- response speed
  Example: - If a student makes many mistakes in a test → workload may be **too high**.

C. Subjective Methods

People **report their own mental effort**.
Example tools:

- questionnaires
- rating scales (NASA-TLX workload scale)

Example question:

> "How mentally difficult was this task?"

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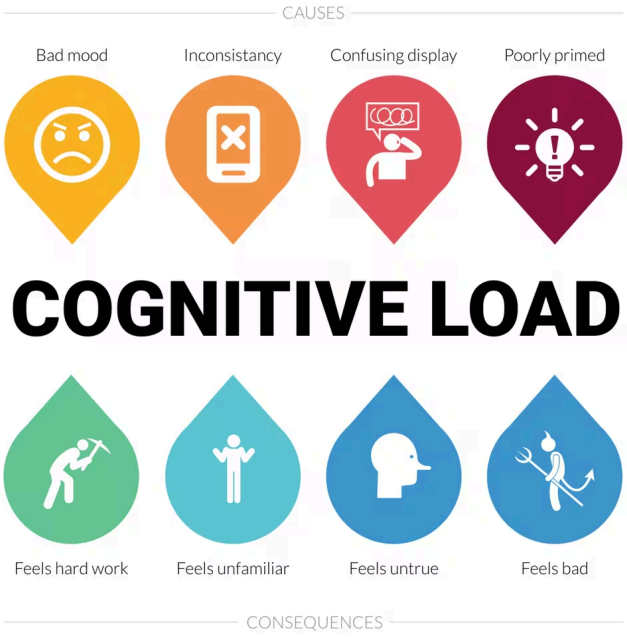
7. Mitigating Cognitive Workload



Mitigating cognitive workload means **reducing the mental effort required to complete a task**.
In simple words:

> **Making tasks easier for the brain.**

Methods to Reduce Cognitive Workload

A. Simplify Information

Provide **clear and simple instructions**.
Example: - Simple app interface instead of complex menus.

B. Reduce Multitasking

Too many tasks at once increase workload.
Example: - Driving + texting → very high cognitive load.

C. Use Visual Aids

Images and diagrams help people understand information faster.
Example: - Charts instead of long text.

D. Automation

Computers perform repetitive tasks.
Example: - Auto spell-check in word processors.